2025

Data Engineering Project- Airline on-time performance analysis

CIS 9760- BIG DATA TECHNOLOGIES

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Milestone 1

Proposal for Airline Data Analysis

Dataset Description
The dataset contains records of airline performance, including details on flight schedules, delays, cancellations, and other operational metrics. This dataset can be leveraged to analyze trends in airline efficiency, punctuality, and factors influencing delays and cancellations.

Data Source
The dataset is available for download at the following location: Airline on-time performance analysis

Dataset Attributes
The dataset includes 29 columns that capture relevant flight details. Attributes are:

1. **Year** – 1987-2008
2. **Month** – 1-12
3. **DayofMonth** – 1-31
4. **DayofWeek** – 1(Monday) – 7(Sunday)
5. **DepTime** – actual depature time (local, hhmm)
6. **CRSDepTime** – scheduled departure time (local, hhmm)
7. **ArrTime** – actual arrival time (local, hhmm)
8. **CRSArrTime** – scheduled arrival time (local,hhmm)
9. **UniqueCarrier** – unique carrier code
10. **Flight Number** – flight number
11. **Tailnum** – plane tail number
12. **ActualElapsedTime** – in minutes
13. **CRSElapsedTime** – in minutes
14. **AirTime** – in minutes
15. **ArrDelay** – arrival delay, in minutes
16. **DepDelay** – departure delay, in minutes
17. **Origin** – origin IATA airport code
18. **Dest** – destination IATA airport code
19. **Distance** – in miles
20. **TaxiIn** – tax in time, in minutes
21. **TaxiOut** – taxi out time in minutes
22. **Cancelled** – was the flight cancelled
23. **CancellationCode** – reason for cancellation (A = carrier, B = weather, C = NAS, D = security)
24. **Diverted** – 1 = yes, 0 = no
25. **CarrierDelay** – in minutes
26. **WeatherDelay** – in minutes
27. **NasDelay** – in minutes
28. **SecurityDelay** – in minutes
29. **LateAircraftDelay** – in minutes

Prediction Objective:

The goal of this analysis is to predict flight delays using machine learning techniques. Based on the nature of the data and the objectives, **logistic regression** will be used to classify flights as "On Time" or "Delayed" based on whether the delay exceeds a certain percentage (X%) of the total trip time. This method is suitable for categorizing flights effectively while allowing for different delay thresholds (e.g., 10%, 25%, 50%).

Machine Learning Approach

By implement logistic regression to predict whether a delay exceeds a given percentage of total trip time. Additionally, we will experiment with different threshold values (10%, 25%, and 50%) to assess how they impact classification accuracy. Other models such as decision trees or random forests may also be explored for improved performance.

In conclusion, this analysis will provide valuable insights into flight delays, helping airlines and passengers make more informed travel decisions. By identifying key contributing factors to develop better strategies to minimize disruptions in air travel.

Milestone 2

Summary

The data acquisition and storage process involved collecting, storing, and organizing data for further processing. The first step was fetching the dataset from **Kaggle** using a shell script with the curl command. The dataset archive (airline-data.zip) was downloaded directly from the Kaggle repository. After downloading, the dataset was extracted using the unzip command. The extracted files included airline.csv.shuffle and carriers.csv, which were required for analysis.

A **Google Cloud Storage (GCS) bucket** named my-bigdata-project-en was created to store the dataset. The bucket was configured with **standard storage class** and **uniform bucket-level access** to manage permissions and access. Authentication was completed using the gcloud auth login command, allowing interaction with the storage bucket. The extracted dataset files were uploaded to the /landing folder in the GCS bucket using the gcloud storage cp command. A verification step was performed to confirm that the files were successfully transferred to the GCS bucket. The contents of the bucket were listed using gcloud commands.

Additional empty folders were created in the GCS bucket for data processing. These folders included **cleaned**, **trusted**, **code**, and **models**. Empty placeholder files were uploaded to these folders using the gcloud storage cp /dev/null command. These steps established a structured process for data acquisition and storage, ensuring the availability of raw data for further analysis. The structured storage allowed for efficient data processing and integration into subsequent workflows.

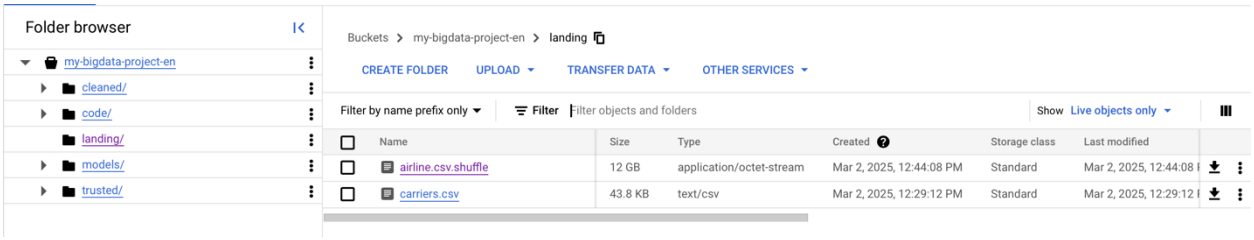
Appendix A Data Acquisition Script

Bucket name:my-bigdata-project-en

1. Fetch the archive file from Kaggle

```
#!/bin/bash

curl -L -o airline-data.zip
https://www.kaggle.com/api/v1/datasets/download/bulter22/airline-data
```



2. Unzip the archive file

```
unzip airline-data.zip
```

```
Processing triggers for man-db (2.11.2-2) ...
nyasembievelyn@instance-20250209-0001:~$ unzip airline-data.zip
Archive:  airline-data.zip
  inflating: airline.csv.shuffle
ls -l

  inflating: carriers.csv
nyasembievelyn@instance-20250209-0001:~$
nyasembievelyn@instance-20250209-0001:~$ ls -l
total 18081312
-rw-r--r-- 1 nyasembievelyn nyasembievelyn  4514927409 Mar  2 16:57 airline-data.zip
-rw-r--r-- 1 nyasembievelyn nyasembievelyn 12029207752 Dec 13  2020 airline.csv.shuffle
-rw-r--r-- 1 nyasembievelyn nyasembievelyn  1971061630 Feb 10 00:16 amazon_reviews_us_Apparel_v1_00.tsv
-rw-r--r-- 1 nyasembievelyn nyasembievelyn       43758 Dec 13  2020 carriers.csv
```

3. Create a bucket on Google Cloud Storage

```
gcloud storage buckets create gs://my-bigdata-project-en --
project= evelyn-nyasembi-project-1 --default-storage-
class=STANDARD --location=us-central1 --uniform-bucket-level-
access
```

4. Using gcloud auth login to authenticate

```
nyasembievelyn@instance-20250209-0001:~$ gcloud storage buckets create gs://my-bigdata-project-en  --project=evelyn-nyasembi-project-1 --default-storage-class=STANDARD --location=us-central1 --uniform-bucket-level-access
Creating gs://my-bigdata-project-en/...
ERROR: (gcloud.storage.buckets.create) HTTPError 403: Access denied. This command is authenticated as 368204563733-compute@developer.gserviceaccount.com which is the active account specified by the [core/account] property.
nyasembievelyn@instance-20250209-0001:~$ gcloud auth login

You are running on a Google Compute Engine virtual machine.
It is recommended that you use service accounts for authentication.

You can run:

  $ gcloud config set account `ACCOUNT`

to switch accounts if necessary.

Your credentials may be visible to others with access to this
virtual machine. Are you sure you want to authenticate with
your personal account?

Do you want to continue (Y/n)?  Y

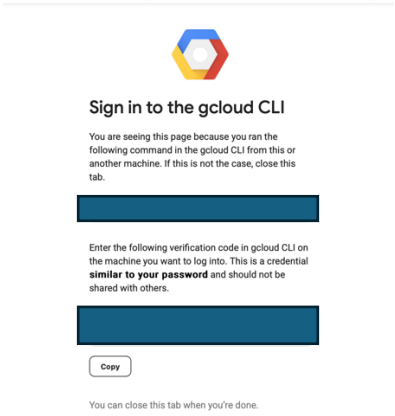
Go to the following link in your browser, and complete the sign-in prompts:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcode.html&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth&state=iZNVOUHL6Fvs7jsDV6vqQyDuEWpyc&prompt=consent&token_usage=remote&access_type=offline&code_challenge=pLuRRW03FLflQFR8ewb2hPZ5X3HyBUcmBptP2j4Xa80&code_challenge_method=S256

Once finished, enter the verification code provided in your browser: 4/0AQSTgQHyf6Rl1M2lxCKH3oZ_fiA9N2CDR69ZMZuxQ_wzyLIR4u41Y85IDUvQgMpfp14H3w

You are now logged in as [nyasembievelyn@gmail.com].
Your current project is [evelyn-nyasembi-project-1].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID


To take a quick anonymous survey, run:
  $ gcloud survey
```



5. Copy the files to GCS bucket in the /landing folder

gcloud storage cp airline.csv.shuffle gs://my-bigdata-project-en/landing/

```
nyasembievelyn@instance-20250209-0001:~$ gcloud storage cp airline.csv.shuffle gs://my-bigdata-project-en/landing/
WARNING: Parallel composite upload was turned ON to get the best performance on
uploading large objects. If you would like to opt-out and instead
perform a normal upload, run:
`gcloud config set storage/parallel_composite_upload_enabled False`
If you would like to disable this warning, run:
`gcloud config set storage/parallel_composite_upload_enabled True`
Note that with parallel composite uploads, your object might be
uploaded as a composite object
(https://cloud.google.com/storage/docs/composite-objects), which means
that any user who downloads your object will need to use crc32c
checksums to verify data integrity. gcloud storage is capable of
computing crc32c checksums, but this might pose a problem for other
clients.

Copying file://airline.csv.shuffle to gs://my-bigdata-project-en/landing/airline.csv.shuffle
  Completed files 32/1 | 11.2GiB/11.2GiB | 155.5MiB/s

Average throughput: 149.1MiB/s
```

gcloud storage cp carriers.csv gs://my-bigdata-project-en/landing/

```
nyasembievelyn@instance-20250209-0001:~$ gcloud storage cp carriers.csv gs://my-bigdata-project-en/landing/
Copying file://carriers.csv to gs://my-bigdata-project-en/landing/carriers.csv
  Completed files 1/1 | 42.7kiB/42.7kiB
```

6. List files in gcloud storage

```
nyasembievelyn@instance-20250209-0001:~$ gcloud storage ls -l gs://my-bigdata-project-en/landing
12029207752  2025-03-02T17:44:08Z  gs://my-bigdata-project-en/landing/airline.csv.shuffle
     43758  2025-03-02T17:29:12Z  gs://my-bigdata-project-en/landing/carriers.csv
TOTAL: 2 objects, 12029251510 bytes (11.20GiB)
```

7. Create empty folders

```
gcloud storage cp /dev/null gs://my-bigdata-project-en/cleaned
gcloud storage cp /dev/null gs:// my-bigdata-project-en/trusted
gcloud storage cp /dev/null gs:// my-bigdata-project-en/code
gcloud storage cp /dev/null gs:// my-bigdata-project-en/models
```

```
nyasembievelyn@instance-20250209-0001:~$ gcloud storage ls -l gs://my-bigdata-project-en
                                 gs://my-bigdata-project-en/cleaned/
                                 gs://my-bigdata-project-en/code/
                                 gs://my-bigdata-project-en/landing/
                                 gs://my-bigdata-project-en/models/
                                 gs://my-bigdata-project-en/trusted/
```

